Title (Units): **COMP7930 Big Data Analytics (3,2,1)**

Course Aims: To introduce the fundamentals of big data analytics; to learn the techniques and tools for big data analytics; to develop proficiency in quantitative methods for big data analytics; to conduct case studies demonstrating the applications of big data analytics.

Prerequisite: COMP7990 Principles and Practices of Data Analytics

Course Intended Learning Outcomes (CILOs):
Upon successful completion of this course, students should be able to:

No.	Course Intended Learning Outcomes (CILOs)
	Knowledge
1	Identify and distinguish big data analytics applications
2	Describe big data analytics tools
3	Explain big data analytics techniques
4	Present cases involving big data analytics in solving practical problems
	Professional Skill
5	Conduct big data analytics using system tools
6	Suggest appropriate solutions to big data analytics problems

Calendar Description: This course aims to introduce the fundamentals of big data analytics and advanced data analytics techniques and tools. Additionally, case studies demonstrate their potential applications across various domains including business, finance industry, and healthcare.

Teaching and Learning Activities (TLAs):

CILOs	Type of TLA
1-3	Students will learn the concepts and techniques via lectures, in-class discussions, quizzes, and assignments.
4-6	Students will learn the skills via guided laboratories and mini-projects.

Assessment:

No.	Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
1	Continuous assessment	40%	1-6	Continuous assessments are designed to measure how well the students have learned the concepts and techniques in big data analytics as well as the skills for solving real-world big-data problems.
2	Examination	60%	1-4, 6	Final examination questions are designed to assess how students have achieved the intended learning outcomes. Analytical questions will be used to assess the understanding of big data analytics concepts and techniques. Problem solving questions will be used to assess the students' ability in tackling applications in big data analytics.

Assessment Rubrics:

Excellent (A)

- Achieve all the six CILOs, demonstrating a good mastery of both the theoretical and practical aspects of big data analytics
- Have a thorough understanding of big data analytics concepts and techniques, and be able to explain and highlight the key points of these concepts
- Able to conduct big data analytics according to application requirements, using system tools such as MapReduce/Hadoop

	• Able to draw on a variety of relevant knowledge and concepts and apply them to solve big data analytics problems
Good (B)	• Achieve all the six CILOs, demonstrating a good understanding of both the theoretical and practical aspects of big data analytics • Have a good understanding of big data analytics concepts and techniques • Able to conduct big data analytics according to most of the application requirements, using system tools such as MapReduce/Hadoop • Able to make use of relevant knowledge and concepts and apply them to solve big data analytics problems
Satisfactory (C)	• Achieve most of the six CILOs, demonstrating a basic level of understanding of the theoretical and practical aspects of big data analytics • Have a basic understanding of big data analytics concepts and techniques • Able to conduct big data analytics according to some of the application requirements, using system tools such as MapReduce/Hadoop • Able to make use of some relevant knowledge and concepts and apply them to solve big data analytics problems
Fail (F)	• Achieve less than three of the CILOs, and have little understanding of the theoretical and practical aspects of big data analytics • Unable to provide solutions to simple problems which require basic understanding of big data analytics concepts • Unable to conduct big data analytics using system tools • Unable to apply big data analytics concepts and techniques to solve practical problems

Course Content and CILOs Mapping:

Content		**CILO No.**
I	Overview of big data analytics	1-4
II	Technologies and tools for big data analytics	1-5
III	Methodologies for big data analytics	3,4,6
IV	Applications and case studies	1,2,4-6

References:

- Jure Leskovec, Anand Rajaraman and Jeffrey David Ullman, Mining of Massive Datasets, 3rd Edition, Cambridge University Press, 2020.
- Vladimir Shikhman and David Müller, Mathematical Foundations of Big Data Analytics, Springer Gabler, 1st Edition, 2021.
- Bill Franks, Taming The Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with Advanced Analytics, Wiley, 2012.
- Neha Narkhede, Gwen Shapira, Todd Palino, Kafka: The Definitive Guide: Real-Time Data and Stream Processing at Scale, O'Reilly Media, 1st Edition, 2017
- Bill Chambers, Matei Zaharia, Spark: The Definitive Guide: Big Data Processing Made Simple, O'Reilly Media, 1st Edition, 2018
- Emc Education Services, Data Science & Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data, John Wiley & Sons Inc, 1st Edition, 2015
- Arshdeep Bahga, Vijay Madisetti, Big Data Science & Analytics: A Hands-On Approach, 2016

Course Content:

<u>Topic</u>

I. Overview of big data analytics
   A. Big data: volume, velocity, variety, veracity, value
   B. Challenges and opportunities
   C. Application examples (e.g., search engines, social networks, e-commerce, generative AI, etc.)

II. Technologies and tools for big data analytics
   A. Centralized and distributed storage systems for big data
   B. Edge computing and cloud computing for big data
   C. MapReduce/Hadoop, Spark
   D. Advanced data visualization (such as t-SNE visualization)

III. Methodologies for big data analytics
   A. Big data pre-processing: similarity measures, data sampling, big data transformation techniques
   B. Dimensionality reduction and locality sensitive hashing
   C. Clustering of big data: BIRCH and BFR algorithm
   D. Real-time analytics: use cases and applications, data streaming architectures, stream processing, Apache Flink
   E. Graph analytics for complex relationship data: graph data, graph representations, search algorithms

IV. Applications and case studies
   including but not limited to the following.
   A. PageRank for web search ranking
   B. Predictive analytics for disease outbreaks (e.g., COVID-19 tracking)
   C. Trend analysis for sales and marketing
   D. Big data in banking: fraud analytics, customer targeting and segmentation